

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EE-8002 (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Application of Power Electronics to Power Systems
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain surge impedance loading.
b) Discuss Ferranti effect.
2. a) Explain the following:
i) Series compensation
ii) Shunt compensation.
b) Describe flexible AC Transmission system controllers. Enumerate benefits due to FACTS controllers.
3. a) What are the objective of static Var compensator in power system? Explain.
b) Explain clearly what do you mean by compensation of line and discuss briefly different methods of compensation.
4. What is a STATCOM? What are its applications? Explain its operation.
5. a) Describe the basic concepts of controlled series compensator.
b) Explain the operation of TCSC with the help of suitable diagram.
6. What is Unified Power Flow Controller (UPFC). Explain its principle of operation, characteristics along with control its performance.
7. Explain Interline Power Flow Controller (IPFC) along with circuit diagram and steady state characteristic. Discuss the advantages of IPFC over UPFC.
8. Explain the following:
i) Compare between HVAC and HVDC transmission.
ii) D-STATCOM
iii) Hybrid active filters
iv) Power Quality problems
